Xin Na (纳鑫)

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Education

Tsinghua University, School of Software, Ph.D. Student 2020.8 - Present

- Advised by Prof. Yuan He.

Tsinghua University, School of Software, Bachelor of Engineering 2016.8 - 2020.7

- With honor: **Excellent Graduate** in Tsinghua University and **Outstanding Graduate** in School of Software.

Research Interests

My research interests lie in wireless networking and ubiquitous IoT, especially battery-free and low-power IoT, cross-technology communication, and wireless sensing. I strive to develop high-performance and reliable IoT systems that enable continuous, ubiquitous sensing across diverse environments. To achieve this vision of sensing *anywhere*, *anytime*, I focus on battery-free and low-power solutions, which I believe hold the greatest promise for scalable and sustainable deployment. My current research in pursuit of this goal includes:

1. Analog backscatter modulation of sensed data without microcontrollers.
2. Parallel RFID communication in the frequency domain via frequency-selective antennas.
3. Truly ubiquitous battery-free backscatter IoT leveraging ambient cellular signals.

Publications

[1] Analog Backscatter for Commodity WiFi with Payload Transparency
Xin Na, Yuan He, Xiuzhen Guo, Jia Zhang, Yang Zou, Zihao Yu, Yunhao Liu.
IEEE/ACM Transactions on Networking (ToN), 2025.

[2] QuinID: Enabling FDMA-Based Fully Parallel RFID with Frequency-Selective Antenna
Xin Na, Jia Zhang, Jiacheng Zhang, Xiuzhen Guo, Yang Zou, Meng Jin, Yimiao Sun, Yunhao Liu, Yuan He.
ACM International Conference on Mobile Computing and Networking (MobiCom), 2025. [PDF] [Code]

[3] Leggiero: Analog WiFi Backscatter with Payload Transparency
Xin Na, Xiuzhen Guo, Zihao Yu, Jia Zhang, Yuan He, Yunhao Liu.
ACM International Conference on Mobile Systems, Applications and Services (MobiSys), 2023. [PDF] [Code] [Slides]

[4] Wi-attack: Cross-technology Impersonation Attack against iBeacon Services
Xin Na, Xiuzhen Guo, Yuan He, Rui Xi.
IEEE International Conference on Sensing, Communication, and Networking (SECON), 2021. [PDF] [Code][Slides]

[5] Satori: In-band Analog Backscatter for Audio Transmission
Yang Zou, **Xin Na**, Yimiao Sun, Yande Chen, Yuan He.
ACM International Conference on Mobile Systems, Applications and Services (MobiSys), 2025. [PDF] (**Best Paper Nominee, 2 out of 233.**)

[6] SmarTiSCH: An Interference-Aware Engine for IEEE 802.15.4e-based Networks
Zihao Yu, **Xin Na**, Carlo Alberto Boano, Yuan He*, Xiuzhen Guo, Pengyu Li, Meng Jin.
ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN), 2022. [PDF]

[7] Trident: Interference Avoidance in Multi-reader Backscatter Network via Frequency-space Division
Yang Zou, **Xin Na**, Xiuzhen Guo, Yimiao Sun, Yuan He.
IEEE International Conference on Computer Communications (INFOCOM), 2024. [PDF]

[8] mmHawkeye: Passive UAV Detection with a COTS mmWave Radar
Jia Zhang, **Xin Na**, Rui Xi, Yimiao Sun, Yuan He.
IEEE International Conference on Sensing, Communication, and Networking (SECON), 2023. [PDF]

[9] Trident: Interference Avoidance in Multi-reader Backscatter Network via Frequency-space Division
Yang Zou, **Xin Na**, Yimiao Sun, Yuan He.
IEEE/ACM Transactions on Networking (ToN), 2024. [PDF]

[10] BIFROST: Reinventing WiFi Signals Based on Dispersion Effect for Accurate Indoor Localization
Yimiao Sun, Yuan He, Jiacheng Zhang, **Xin Na**, Yande Chen, Weiguo Wang, Xiuzhen Guo.
ACM Conference on Embedded Networked Sensor Systems (SenSys), 2023. [PDF]

[11] Detection and Identification of Non-cooperative UAV Using a COTS mmWave Radar
Yuan He, Jia Zhang, Rui Xi, **Xin Na**, Yimiao Sun, Beibei Li.
ACM Transactions on Sensor Networks (TOSN), 2024.[PDF]

[12] RFinder: Pinpoint the Invisible RFID Tags in the Prefabricated Buildings
Jiacheng Zhang, Meng Jin, Yimiao Sun, Weiguo Wang, Jia Zhang, **Xin Na**, Xiuzhen Guo, Yuan He.
International Conference on Embedded Wireless Systems and Networks (EWSN), 2024.[PDF]

[13] Real-Time Continuous Activity Recognition With a Commercial mmWave Radar
Yunhao Liu, Jia Zhang, Yande Chen, Weiguo Wang, Songzhou Yang, **Xin Na**, Yimiao Sun, Yuan He.
IEEE Transactions on Mobile Computing (TMC), 2024.[PDF]

[14] Cross-Technology Communication for the Internet of Things: A Survey
Yuan He, Xiuzhen Guo, Xiaolong Zheng, Zihao Yu, Jia Zhang, Haotian Jiang, **Xin Na**, Jiacheng Zhang.
ACM Computing Surveys (CSUR), 2022.[PDF]

[15] A Survey of mmWave-Based Human Sensing: Technology, Platforms and Applications
Jia Zhang, Rui Xi, Yuan He, Yimiao Sun, Xiuzhen Guo, Weiguo Wang, **Xin Na**, Yunhao Liu, Zhenguo Shi, Tao Gu.
IEEE Communications Surveys & Tutorials (COMST), 2023.[PDF]

Tools and Open-Source Researches

Beyond research, I also dedicate myself to building and sharing tools, hardware, and software to advance the wireless networking community.

QuinID, hardware and software for an FDMA-based fully parallel RFID system.

- The UHF RFID band is divided into five sub-bands, resulting in a fivefold improvement in reading rate.
- An additional tool, UHF-Gen2-RFID-Reader, is provided—a high-performance SDR-based reader for the broad RFID research community, achieving up to 1000 reads per second.

Leggiero, the entire hardware of an analog WiFi backscatter.

- The tag converts the analog sensor voltage to the RF signal phase and embeds it in the CSI.
- An updated version, Leggiero+, is also provided with a few enhancements made to the circuit design.

Wi-attack, the software of a cross-technology impersonation attack method against iBeacon.

- Uses WiFi devices to attack iBeacon services by impersonating an iBeacon base station.

Professional Activities

ACM Transactions on Internet of Things, Reviewer	2025
IEEE Transactions on Mobile Computing, Reviewer	2024
Contributed to paper reviews for INFOCOM, IPSN, SenSys, SECON	2021 - Now
Teaching Assistant, Network Measurement and Analysis, Instructor: Prof. Yuan He	2021 - 2024

Honors and Awards

Talents Scholarship *(Top 15%)*, Tsinghua University	2021 - 2024
Excellent University Graduate *(Top 10%)*, Tsinghua University	2020.6
Outstanding Graduate *(Top 10%)*, School of Software	2020.6
National Encouragement Scholarship *(Top 5%)*, Tsinghua University	2018, 2019